Hazem Alsagheer . Jarvis Consulting

I am a Computer Engineering graduate from Toronto Metropolitan University with experience building data-driven systems at the intersection of analytics and infrastructure. During my time at Hydro One, I developed Python and SQL pipelines to process and validate enterprise data, identifying critical inconsistencies that helped prevent potential regulatory losses. I have also worked in a research setting applying topic modeling techniques to large-scale datasets, where I focused on improving clustering quality through better model evaluation strategies.

What excites me about the software industry is building systems that actually work at scale and have real impact. Im especially interested in the data side, where even small improvements in data quality or pipeline design can make a big difference in decision-making. I enjoy understanding how data flows through a system, spotting where things break down, and building solutions that make it more reliable and efficient.

Skills

Proficient: Python, RDBMS/SQL (PostgreSQL, MS SQL Server), Data Modeling (Normalization, ERD Design), ETL Pipeline, Data Analysis (pandas, NumPy), FastAPI, Data Structres & Algorithms, Git

Competent: Docker, AWS (EC2, deployment), Machine Learning, JavaScript, TypeScript, React.js, Java

Familiar: Linux/Bash, C/C++, GCP, VBA, Networking

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_HazemAlsagheer

Linux Cluster Monitoring System [GitHub]: Built a lightweight Linux cluster monitoring system where each node collects hardware specifications and real-time resource metrics and sends them to a centralized PostgreSQL database. Designed the system to give administrators clear visibility into CPU, memory, and disk usage for better resource allocation and capacity planning. Implemented a Bash-based agent using tools like vmstat and df, with data ingestion handled through a containerized database deployed via Docker. Automated data collection with cron and validated accuracy through script testing and SQL-based analysis.

Java Grep App [GitHub]: Implemented a Java-based Grep application that recursively scans directory structures and extracts file content matching user-defined regular expression patterns, mimicking core Linux grep functionality. Developed two separate implementations using both imperative programming and Java Streams/lambda expressions to compare readability, maintainability, and performance tradeoffs under different workloads. Utilized Maven for dependency management and packaging, JUnit for automated testing, SLF4J/Logback for application logging, and Docker for containerization and deployment portability.

Highlighted Projects

Journaling app: Echos of Ink [GitHub]: Designed and deployed a full-stack journaling application with a focus on reliable data flow and secure system architecture. Built a FastAPI backend with JWT-based authentication and refresh token rotation, paired with a React TypeScript frontend using Axios interceptors to manage concurrent requests and session state. Modeled and managed relational data in PostgreSQL, including token lifecycle handling for security and consistency. Containerized the application with Docker and deployed it on AWS EC2 behind an Nginx reverse proxy with HTTPS, ensuring scalable and production-like behavior across environments.

Capstone Project: Biomedical Quesiton Answering Model: Built a biomedical question-answering system by fine-tuning BioBERT on 3K+ BioASQ list-type questions, improving F1-score from 3% to 53.7% through a custom multi-span extraction strategy. Designed a post-processing pipeline to rank and filter top-N predictions, reducing redundancy and improving answer precision. Focused on optimizing the end-to-end data and inference pipeline to ensure accurate, structured outputs for domain-specific queries.

Professional Experiences

Data Engineer, Jarvis (2026-present): Working in a fast-paced, sprint-driven engineering environment focused on building scalable software and data-driven solutions across multiple technologies and domains. Collaborate with peers through daily standups, technical discussions, and project reviews while consistently delivering projects under tight deadlines. Regularly apply problem-solving, debugging, and system design skills in an iterative development setting, adapting quickly to new tools, frameworks, and engineering concepts while communicating technical decisions clearly.

Data Analyst, Hydro One Inc. (2023-2024): Designed an SQL and Python-based ETL pipeline to spot and resolve data inconsistencies across enterprise systems, saving the company a potential regulatory fine of $1M/day. Wrote complex SQL queries to analyze large datasets and generate actionable insights, enabling stakeholders to make faster, data-driven decisions. Resolved data quality issues at the source across cross-functional systems, improving overall data reliability by 66%. Mentored incoming co-op students on SQL development, system workflows, and best practices, strengthening team efficiency.

Research Assistant, Toronto Metropolitan University:Cybersecurity Research Lab (2024-2025): Conducted end-to-end analysis of a BERTopic pipeline on large-scale cybersecurity corpora, evaluating embedding performance and topic coherence. Identified inefficiencies in the default topic selection process, which relied on static parameter choices and led to suboptimal clustering. Designed an interval-based optimization strategy to iteratively evaluate topic counts and cluster configurations, improving coherence and interpretability. Presented findings to senior researchers, contributing to refinement of the modeling approach.

Education

Toronto Metropolitan University (2020-2025), Bachelors of Engineering, Electrical and Computer Engineering - Dean's List (2020 - 2025) - GPA: 4.06/4.33

Miscellaneous

- Piano
- Chess
- RockClimbing
- Calisthenics
- Muay Thai
- Soccer
- Basketball